

Processor Design Project (CESE4040)

Grading Scheme

Q4, 2025–2026

1. Overview

The final grade for every student consists of two components:

Component	Description
Group grade	Determined by the quality of the project as a whole, assessed during the final examination (presentation, demo, Q&A, and submitted artefacts).
Individual adjustment	Applied on top of the group grade based on (a) individual Q&A performance during the final examination and (b) peer evaluation via Buddy Check. Range: ± 1 point.

Final individual grade = Group grade + Individual adjustment, clamped to $[1, 10]$ and rounded to the nearest half point.

A minimum final individual grade of **6.0** is required to pass the course. Buddy check is applied only when the group grade is 6 or above.

2. Mandatory Requirements

Before the rubric is applied, the following mandatory deliverables are checked. **All** of the following must be functional and demonstrated during the final examination.

1. The baseline system runs: software AES encryption executes correctly on the unmodified RISCY core (simulation and/or FPGA).
2. A profiling/bottleneck analysis of the baseline system has been performed and documented.
3. The RISC-V scalar cryptography instructions `aes32esi` and `aes32esmi` are implemented in the RISCY core and produce correct encryption output.
4. The LLVM built-in loop-unrolling pass is applied and the unrolled code produces correct output.

A group that meets all four requirements receives a **base group grade of 6**. The rubric in Section 3 then determines the grade above 6, based on the quality and scope of the work beyond these mandatory requirements.

A group that **fails** to meet one or more mandatory requirements receives a **group grade of 5 (fail)**. The group grade is applied equally to all members and no individual adjustment (Q&A or Buddy Check) is considered.

3. Group Grade Rubric

The rubric evaluates the work beyond the mandatory requirements: the proposed improvement(s), their implementation, the results and analysis, and the final presentation. Each criterion is scored on a four-level scale. The criteria and their weights are:

#	Criterion	Weight
C1	Suitability of the proposed solution	20%
C2	Quality of implementation	25%
C3	Depth of results and analysis	30%
C4	Presentation, demo, and Q&A	25%

Scoring

Each criterion receives a score:

Level	Points
Excellent	4
Good	3
Satisfactory	2
Unsatisfactory	1

The weighted average is computed and mapped to the group grade:

$$\text{Group grade} = 6 + (W - 1) \times (4 / 3)$$

where $W = 0.20 \times C1 + 0.25 \times C2 + 0.30 \times C3 + 0.25 \times C4$. This produces the following reference points:

All criteria at level	Weighted avg (W)	Group grade
Unsatisfactory	1.0	6.0
Satisfactory	2.0	7.3
Good	3.0	8.7
Excellent	4.0	10.0

The computed grade is rounded to the nearest half point.

Note: a group that passes the mandatory gate but scores Unsatisfactory on all four rubric criteria receives a group grade of **6.0**. To score above 6, at least one criterion must be above Unsatisfactory.

C1 — Suitability of the Proposed Solution (20%)

This criterion evaluates whether the group's chosen improvements are well-motivated, relevant to the identified bottlenecks, and grounded in the state of the art. Each level builds on the previous one.

Level	Description
Unsatisfactory	The chosen improvement(s) appear arbitrary or disconnected from any baseline analysis. No clear target metric or hypothesis is formulated. Little or no reference to related work.
Satisfactory	The improvement(s) are relevant to the system and a target metric is defined. A baseline analysis has been performed and the improvement idea is connected to it, even if the justification is shallow. Some related work is mentioned but not discussed in depth.
Good	Builds on Satisfactory: the connection between baseline bottleneck(s) and the proposed improvement(s) is clearly argued. The state of the art is surveyed with reasonable depth and the group positions their approach relative to existing solutions. The improvement hypothesis is stated and the evaluation methodology is outlined.
Excellent	Builds on Good: the group demonstrates a thorough and critical understanding of the state of the art, going beyond listing papers to comparing approaches and justifying their specific choice. The hypothesis is specific, testable, and quantified where possible. The choice of improvement shows originality or particularly strong technical insight.

C2 — Quality of Implementation (25%)

This criterion evaluates the technical quality of the hardware (RTL) and software (LLVM/C) implementation beyond the mandatory requirements. Correctness of the mandatory part is a prerequisite (see Section 2), not a distinguishing feature. Each level builds on the previous one.

Level	Description
Unsatisfactory	No meaningful extension beyond the mandatory part, or the extension(s) are non-functional. The submitted code is disorganized or poorly understood by the group members themselves.
Satisfactory	At least one extension is attempted and partially functional. The implementation shows that the group made a genuine effort, even if the result has limitations or bugs that are acknowledged. The code is readable and the group can explain their approach.
Good	Builds on Satisfactory: the extension(s) are functional and produce correct results. The code is reasonably structured with meaningful signal/variable naming and comments where needed. Limitations are identified and documented rather than hidden.
Excellent	Builds on Good: the implementation is non-trivial in scope or complexity and demonstrates strong engineering practice: clean modular structure, reusable components, and evidence of systematic verification (e.g., dedicated testbenches, test vectors). Edge cases and failure modes have been identified and handled. The group shows mastery of the tools (Vivado, LLVM) beyond basic usage. The approach is eligible for publication.

C3 — Depth of Results and Analysis (30%)

This criterion carries the highest weight. It evaluates the experimental methodology, the quality of the results, and critically the group's ability to reason about what they observe, including unexpected or negative results. Each level builds on the previous one.

Level	Description
Unsatisfactory	Results are incomplete or missing key metrics. No meaningful comparison against a baseline. The group cannot explain their own results or draws unsupported conclusions.
Satisfactory	A baseline is defined and results are presented with at least one relevant metric (e.g., clock cycles, resource utilization). The methodology is described, even if it has weaknesses. The analysis is mostly descriptive; the group reports what they observed but offers limited reasoning about why. Negative results, if any, are reported but not explained.
Good	Builds on Satisfactory: multiple relevant metrics are reported and the methodology is sound and clearly described. The group provides reasonable interpretation of the results, connecting observations back to the design decisions. Comparisons against the baseline are fair and quantitative. Minor gaps in reasoning or missing secondary comparisons.
Excellent	Builds on Good: the group provides insightful analysis explaining why results are as observed, not just confirming improvement but reasoning about its source (e.g., which pipeline stage, which memory access pattern). Trade-offs are explicitly discussed (e.g., area vs. speed, code size vs. latency). If an idea did not yield the expected improvement, the group provides a well-reasoned explanation grounded in evidence. Results are compared against state-of-the-art references where applicable. The methodology is reproducible by an independent reader. The approach is eligible for publication.

Note: A well-analyzed negative result (an improvement that did not work, with a clear explanation of why) is valued higher than a positive result with no analysis. The goal is to demonstrate understanding, not to guarantee speedup.

C4 — Presentation, Demo, and Q&A (25%)

This criterion evaluates the final examination session: the 20-minute presentation, the 10-minute live demo, and the 20-minute Q&A. Each level builds on the previous one.

Level	Description
Unsatisfactory	The presentation is disorganized or missing required topics. Technical content contains significant inaccuracies. The demo fails or is absent. Q&A reveals fundamental gaps in understanding among multiple group members.
Satisfactory	All required topics are covered and the presentation is understandable, even if the structure or flow could be improved. The demo runs and shows the system working. All group members present, though some may be noticeably less prepared. Q&A responses are mostly correct but sometimes vague.
Good	Builds on Satisfactory: the presentation is well-structured with a clear narrative from problem to results. Slides are visually clean and support (rather than repeat) the spoken content. The demo is convincing and runs smoothly. All group members present competently. Q&A responses are accurate and show understanding beyond the slides.
Excellent	Builds on Good: the presentation demonstrates strong technical communication: complex ideas are explained clearly, design decisions are justified rather than just described, and the group anticipates likely questions in their narrative. The demo is well-prepared with clear before/after comparisons. All group members show confident command of the full project, not just their own part. Q&A responses show depth of understanding and the ability to reason on the spot about hypothetical variations or limitations.

Required Presentation Topics

The 20-minute presentation must cover at least:

1. Problem definition
2. Design overview
3. Implementation details
4. Results and discussion/analysis (linked to metrics defined in the intermediate report)
5. Potential future work
6. Short reflection on group performance during the project

Slides must be submitted on Brightspace **one week** before the final examination date. Minor changes after submission are permitted.

4. Individual Adjustment (± 1 Point)

The individual adjustment differentiates students within a group. It is applied **only when the group grade is 6 or above**. If the group grade is below 6 (failed mandatory gate), the group grade applies equally to all members and no individual adjustment is considered.

The adjustment consists of two equally weighted components:

Component	Range	Assessed by
Individual Q&A performance	± 0.5	Examiners
Buddy Check (peer evaluation)	± 0.5	Group members

4.1 Individual Q&A Performance (± 0.5)

During the 20-minute Q&A following the presentation, examiners will direct questions to individual students. The purpose is to assess each student's understanding of (a) the overall project and (b) their own specific contribution.

Score	Description
+0.5	The student demonstrates excellent understanding of the project as a whole and can explain their own contribution in depth. Answers are accurate, confident, and show insight beyond surface-level knowledge.
0	The student demonstrates adequate understanding. Can explain their contribution and has a reasonable grasp of the overall project.
-0.5	The student struggles to explain their own contribution or shows poor understanding of the project. Answers are vague, incorrect, or suggest minimal involvement.

4.2 Buddy Check — Peer Evaluation (± 0.5)

Each student completes a peer evaluation form, rating every group member on three dimensions. Ratings use a 5-point scale (1 = Strongly Disagree, 5 = Strongly Agree).

Communication

The team member communicated proactively and effectively. They shared progress, raised issues early, participated constructively in discussions, and kept the team informed about their work.

- 5 — Consistently kept the team informed, raised issues proactively, and contributed constructively to all discussions.
- 4 — Communicated regularly and effectively, with only minor gaps.
- 3 — Communicated adequately when prompted but rarely took the initiative.
- 2 — Communication was infrequent or unclear, causing coordination issues.
- 1 — Was largely unresponsive or absent from team communication.

Effort and Quality of Work

The team member invested sufficient time and energy into the project. Their deliverables were of good quality, technically sound, and completed on time.

- 5 — Consistently delivered high-quality work on time and often went beyond what was expected.
- 4 — Delivered good-quality work on time, with only minor issues.
- 3 — Delivered acceptable work but occasionally missed deadlines or produced work requiring significant revision.
- 2 — Deliverables were frequently late, incomplete, or of poor quality.
- 1 — Contributed very little usable work to the project.

Commitment and Reliability

The team member was dependable throughout the project. They followed through on commitments, attended meetings and lab sessions, and took responsibility for their tasks.

- 5 — Fully dependable throughout the project; always followed through and was present when needed.
- 4 — Reliable with only occasional lapses.
- 3 — Sometimes missed commitments or meetings but generally made up for it.
- 2 — Frequently unreliable, causing the team to compensate for missed tasks.
- 1 — Rarely followed through on commitments; the team could not depend on this member.

Computation of Buddy Check Adjustment

For each student, the **average peer rating** (excluding self-assessment) across all three dimensions is computed. The group mean of all such averages is then calculated. The individual adjustment is based on how far each student's average deviates from the group mean:

Deviation from group mean	Adjustment
$\geq +1.0$ above group mean	+0.5
Between -1.0 and $+1.0$	0
≤ -1.0 below group mean	-0.5

Self-assessment scores are collected for calibration purposes but are **not** used in the adjustment calculation.

In cases where a student's peer ratings are unanimously very low (average below 2.0), examiners will review the case individually.

5. Summary of Grade Calculation

1. Check mandatory requirements. If any are missing → **group grade = 5 (fail)**, applied to all members equally. No individual adjustment. Stop here.
2. If all mandatory requirements are met → base group grade = 6.
3. Score each of the four rubric criteria (C1–C4) on a 1–4 scale.
4. Compute weighted average: $W = 0.20 \times C1 + 0.25 \times C2 + 0.30 \times C3 + 0.25 \times C4$
5. Compute group grade: $G = 6 + (W - 1) \times (4/3)$, rounded to nearest 0.5.
6. Compute individual adjustment: $A = \text{Q\&A adjustment } (\pm 0.5) + \text{Buddy Check adjustment } (\pm 0.5)$. Buddy Check uses peer ratings only; self-assessment is excluded.
7. Final individual grade: $F = G + A$, clamped to $[1, 10]$, rounded to nearest 0.5. Must be ≥ 6.0 to pass.

6. Important Notes

- An idea that does not produce the expected improvement is **not penalized** as long as the group provides a well-reasoned analysis of why the result differs from the hypothesis. Honest and insightful analysis of negative results is valued higher than superficial reporting of positive results.
- The number of extensions is **not** a grading factor by itself. One thoroughly implemented and analyzed improvement can score higher than multiple shallow ones.